

Practice Quiz 01D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 01

Truth tables and logical equivalence.

1. Use a complete truth table to determine whether

$$(P \vee Q) \wedge (P \vee \neg Q) \quad \text{and} \quad P$$

are logically equivalent. State your conclusion.

Solution

P	Q	$P \vee Q$	$P \vee \neg Q$	$(P \vee Q) \wedge (P \vee \neg Q)$
F	F	F	T	F
F	T	T	F	F
T	F	T	T	T
T	T	T	T	T

The final column agrees with P in every row, so the formulas are logically equivalent.

The distributive law also gives $(P \vee Q) \wedge (P \vee \neg Q) \equiv P \vee (Q \wedge \neg Q) \equiv P$. ■